
SILVERPOINT VACUUM TRUCK SERVICES
STANDARD OPERATING PROCEDURE
Water Truck Loading (COLD/ICE)

Proc. No: 1.41
2024/01/12

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PURPOSE

This Standard Operating Procedure (SOP) is to identify potential hazards and outline the steps for loading the Water Truck in a COLD Environment / Froze Over Water Hole, Ice.

HAZARDS

TYPE	HAZARD	TYPE	HAZARD	TYPE	HAZARD
Physical	Slips, Trips, Falls	Chemical	H2S	Property	Equipment
	Fall from Height		Fuel		Third Party Property
	Pinch Points		Cement	Productivity	Down time
	Heated/Hot Surface	Biological	Wildlife		Inefficient
	Noise 82<		Insect bites	Legal	Suspension
	Energy		Influenza		Out of Service
	Environment	Social	Harassment		Fine
	Confined Space		Violence		
	Work Alone				

REFERENCES

PPE	Prevention	References
Fire Retardant Coveralls	Hazard Assessment	Alberta OH&S
Steel Toed Boots	Inspection	Task List
Safety Glasses	Maintenance	Standard Operating Procedure
Gloves	FLHA	
Hard Hat	Safety Meeting	
Hearing Protection	Ear Plugs / Muffs	
Flotation Jacket	Assigned to truck	
Harness	Assigned to truck	
Lifeline	Assigned to truck	

You are required to load at a Water Hole, there are some items that need to be considered:

Is this your first trip to the Water Hole;	Do you have a Safety Attendant;
Has the Water Hole been Risk Assessed;	Will the Safety Attendant be accompanying you;
How far away is the Water Hole, Travel Time, ERP Response Time;	Will the Safety Attendant be available by cell phone or other;
Do you have a WORK ALONE PLAN in place?	

Water Truck Loading: COLD / ICE Pre-Job

- 1) All these factors should be seen as potential dangers, and each requires a hazard management strategy. Treating a natural water hole differs from handling a man-made one. Regardless, a water hazard poses a significant risk, and a Hazard Management plan must be in place, including a Safety Attendant, a flotation jacket, harness, and lifeline. Take into account the environment, time for help to arrive, and the presence of wildlife.

Refer to SOP 4.02 Work Alone

If you do not have a WORK ALONE PLAN in place.

You do not go to the Water Hole unless you have a WORK ALONE PLAN activated.

- 2) **PARK THE TRUCK:** When you approach the new work site, look for a place to park your unit. You want to look at your best option to position your truck in alignment to the water hole.

Refer to SOP 2.01 Backing Up & Tight Spaces

- 3) **CONTACT SAFETY ATTENDANT:** Work Alone / Safety Attendant;

- a. If this is the first trip to the water hole, a Safety Attendant needs to accompany you. The Safety Attendant and yourself will complete a Field Level Hazard Assessment and review the Work Alone Plan, ensuring that all hazards are addressed.
 - i. If this is your second trip or more and it has been decided that a lower risk has been assigned. You will not need a Safety Attendant to accompany you. You will need to perform pre-arranged Work Alone Check-in Calls.
 - ii. Upon leaving the **Rig Site ensure a Safety Attendant** has been assigned to you. Obtain their Phone number and determine the Frequency of calls and or text messages between you. If the Safety Attendant does not receive a message what amount of time should they wait before activating the Emergency Response Plan.

Refer to SOP 4.02 Working Alone

- 4) **IDENTIFY THE WATER HOLE:** Determine the water hole and how it is situated, is it a natural water hole? Or is it man made with a dyke/wall? Is it lined? Or is it another set up or type of water hole? Will you be working within **TWO METERS** of the water hole embankment? Base your Tasks and steps on where and the type of water hole you will be dealing with.

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- 5) **PERFORM FLHA, OBTAIN ICE THICKNESS:** If this is your first trip to the water hole? Start your Field Level Hazard Assessment Form (FLHA), noting all the Hazards and how you will manage each one.
- a. Are you working within **TWO METERS** of the Water Hole embankment?
 - b. Flotation Jacket, Harness, and Lifeline must be used.
 - c. **What is the ICE Thickness?** Have you been informed of the Ice Thickness?
Do you have to determine the Ice Thickness your self?
 - d. You may have to drill a hole through the ice to determine what the Ice Thickness is.
 - e. It is COLD, the water surface is frozen. How thick is the ice? Do not step on to the ice unless the ice has been verified as safe. Without measuring the thickness, you will not know. The following table will provide an idea of what ICE thickness allows for safe work. (Alberta Working on ICE)

Minimum Ice Thickness for Stationary/Parked loads up to 5,000 Kg (For more than Two hours, but less than Seven days)	
Load/Situation	Minimum Effective Ice Thickness (cm)
Person Standing	15cm
ATV or Snowmobiles (Maximum Weight machine + rider <500 kg)	25cm

- d. Upon Completion, the FLHA needs to be signed by your self and your Safety Attendant.

6) **STANDARD PPE APPLIES ADD FLOATION JACKET, HARNESS, and LIFELINE.**

Wear your PPE and add the PPE required for working around WATER, PPE must be worn. You must implement the Harness and Lifeline if you are working within TWO METERS of the Water Hole embankment. Inspect your PPE and if there are defects such as; Cuts, Tears or other you cannot go near water.

Flotation Jacket must be worn over top of all outerwear and the harness.
(Flotation Jacket will not inflate properly if restricted by other clothing PPE.)

COLD WEATHER LOADING

- 7) **POSITION THE TRUCK FOR LOADING:** Select the best position adjacent to the water hole, about 2m away from the water hole embankment, keeping in mind less hose is better, with cautions. Not enough hose can be a hazard that would also hinder the water loading operation. To many hoses, makes for more chance on leaking connections or air preventing suction or prime. With less curves in the hose and making them more strait, the better it will be for walking out the hose and suction. Position the truck where you think it is best, use the Safety Attendant on the first trip to help park and then mark your parking position for additional trips.

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ATTENTION: Wear your Safety Equipment PPE, Use your Harness / Lifeline and Floatation Jacket when you are with TWO METERS of waters Edge, Is your Safety Attendant with you (first trip) or do you have contact with a Safety Attendant is a Work Alone Plan in place?

Refer to SOP 2.01 Backing Up & Tight Spaces

- 8) **CONNECT A HOSE TO THE TRUCK PUMP INLET:** Ensure of a good connect, make sure the Cam Locks are secure.

Refer to SOP 1.40 Water Truck Operation, SOP 1.02 Connecting & Disconnecting Hoses

- 9) **CONNECT HARNESS TO LIFELINE AND SECURE LIFELINE TO TIE OFF POINT:** The harness and lifeline are exactly how they are titled; the Lifeline attached between your harness and an anchor. The assembly is what will get you back to solid ground should you break through the ice and fall into the water.

a. **ANCHOR:** On the back end of the truck there is a Tie Off Point, where you secure your lifeline to. Make sure you have enough length to perform your work. But short enough that the lifeline is not in your way of work. If you get into trouble, you can pull your self to safety.

b. **Safety Attendant:** if a Safety Attendant is accompanying you, the lifeline still needs to be secured to the tie off point. If you get into trouble the Safety Attendant will assist pulling you to safety.

i. **The Safety Attendant:** will feed the lifeline out until the operator reaches the Water Hole. Then the Safety Attendant will Tie Off to the Tie Off Point.

- 10) **DETERMINE ICE THICKNESS:** You may be fortunate enough to have the Ice Thickness determined in advance for you. If you do not, you will have to determine Ice thickness on your own. Use the Table in Step 5 and

Refer to SOP 4.07 Working on Ice, SOP 4.02 Working Alone

- 11) **IDENTIFY LOCATION TO FIND OR CREATE HOLE IN ICE:** Determine the best location to drill and create a hole through the ice. Use a hand auger be sure to follow the Working on Ice SOP 4.07, using the steps for the type of Auger that you are using.

If you are going to use a bar to break the ice, do this, as with any type of tool use extreme caution.

Refer to SOP 4.07 Working on Ice, SOP 4.02 Working Alone

- 12) **PLACE COVER OVER HOLE IN ICE:** Place a cover, to cover the hole in the ice helping to prevent your water hole from freezing over.

IE: Sawdust bag or a piece of plywood, remember nothing can be left on the ice after you are finished hauling water from this water source, must be cleaned up and disposed of properly.

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- 13) **ASSEMBLE THE HOSES REQUIRED:** determine what you will need for hose length to go from the Water Truck to the water hole. Remove the hose (s) and lay the hoses end to end. Follow through connecting each length of hose to the other hose via the couplers, ensuring the Cam Locks are securely fastened with the O-ring rubbers.

Refer to SOP 4.03 Heavy Lifting Hose, SOP 1.02 Connecting and Disconnecting Hoses

- 14) **INSTALL SCREEN ON THE END OF THE SUCTION HOSE:** The screen is important to ensure the debris or fish does not plug up your intake and or get into the pump and cause damage.
Install the proper sized screen.

Refer to SOP 1.40 Water Truck Operation

- 15) **UNCOVER HOLE IN ICE:** Uncover the hole in the ice.

Refer to SOP 4.07 Working on Ice, SOP 4.02 Working Alone

- 16) **PLACE THE SUCTION END OF THE HOSE INTO THE HOLE IN THE ICE TO REACH THE WATER:**

Avoid stepping onto the ice, if possible, slide the suction end of the hose to the hole in the ice and attempt to get the end of the hose into the hole. If this is not possible, ensure your harness and lifeline are secured to an anchor point. Then proceed and step out onto the ice.

Refer to: SOP 4.07 Working on Ice, SOP 4.02 Working Alone, SOP 1.40 Water Truck Operation

- 17) **OPEN THE VALVE ON THE SUCTION INLET, ENGAGE THE PUMP:** Open the Valve on the Suction Inlet and engage the PUMP as per SOP 1.40. Water will start moving from the Water Hole into the Truck Tank. Check your hose connections for leaks. If you have a leak STOP the Pumping and determine why the connection is leaking and make adjustments or repairs.

Refer to SOP 1.40 Water Truck Operation

- 18) **MONITOR AMOUNT OF WATER INTO TANK:** Monitor the amount of water that has been loaded into your truck tank. Take into consideration road weight limits.
WHEN LOADED, STOP LOADING.

Refer to SOP 1.40 Water Truck Operation

DISCONNECTING FROM LOADING OPERATION (TO STOP LOADING)

- 19) **ENSURE HARNESS AND LIFELINE IS CONNECTED AND SECURE TO A TIE OFF POINT: Working within Two Meters of the water EMBANKMENT.** Ensure that the Lifeline is still properly secured to the TIE PFF POINT.

REMEMBER: if you get into trouble this is your way to pull your self out of the water and to SAFETY.

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20) PLACE COVER OVER HOLE IN ICE: Place a cover, to cover the hole in the ice helping to prevent your water hole from freezing over.

- a. **IE: Sawdust bag or a piece of plywood,** remember nothing can be left on the ice after you are finished hauling water from this water source, must be cleaned up and disposed of properly.

Refer to SOP 4.07 Working On Ice, SOP 4.02 Working Alone

21) WALK OUT REMAINING HOSE (S) TO PUMP TO DRAIN WATER. With pump engaged Walk out hose (S) to drain water all the way back to the truck and into pump.

22) CLOSE VALVE DISENGAGE PUMP: Close the valve on suction inlet and disengage the pump.
Refer to SOP 1.40 Water Tuck Operation, SOP 1.02 Connecting Disconnecting Hoses

23) DISCONNECT LOAD HOSE FROM THE SUCTION HOSE. If you will be returning to the water hole for additional loads of water, leave the hose (s) in an orderly manner. The Suction end of the hose can be placed on top of the Sawdust bag over the Ice hole or on top of which ever material you are using to cover the hole.

Refer to SOP 1.40 Water Truck Operation, SOP 1.02 Connecting Disconnecting Hoses

- a. **SAFELY BACK ON SOLID GROUND, TWO METERS FORM WATERS EDGE, REMOVE LIFELINE FROM TIE OFF POINT.** Once you are back into the Water Truck area, with no need to go back to the waters edge you can remove the Lifeline from the tie off point.
- b. **DISCONNECT HOSE (S) PICKUP HOSE (S) AND SECURE TO THE TRUCK:** Inspecting each hose, the couplers, and cam locks. If a hose is found to have a defect be sure to mark the hose and place on the truck for easy removal. Place all other Hose (s) on the truck, secure all the hose (s).
- c. **CHECK AREA, DEAL WITH ANYTHING THAT NEEDS ATTENTION.** Re-check your work site, anything left unattended, a hose left behind, or other tool IE: Ice Auger. Pick up and remove sawdust bags or wood that you used to cover your water source hole and clean up what may be there.

STOW COLD OPERATION SAFETY EQUIPMENT

- 24) REMOVE SAFETY GEAR FLOTATION JACKET, HARNESS & LIFELINE:** Remove the Lifeline, the harness and flotation jacket.
- 25) INSPECT SAFETY GEAR FLOTATION JACKET, HARNESS AND LIFELINE:** Inspect all pieces for abrasions, tears, or other types of damage. If you found damage mark the damaged equipment and request a replacement.
- 26) STOW SAFETY EQUIPMENT; FLOTATION JACKET, HARNESS AND LIFELINE.** Place each in a dry place on the truck. The Lifeline can be coiled into a 5-gal bucket.
- 27) CONTACT SAFETY ATTENDANT:** Contact via a Cell phone or other method Two-way radio? IF contact is not made with your Safety Attendant try the other Site numbers. Failing contact send a text message that you are leaving the water hole and traveling back to SITE.

Refer to SOP 4.02 Work Alone

If you any questions STOP WORK and contact your Supervisor

REGULATIONS

Work Alone	AB OHS Act, Section 2, 35, and 36
	AB OHS Code Alberta Regulation 191/2021 Work Alone Part 28
Right to Refuse	AB OHS Act Part 3 Section 35
Hazard Prevention	CA Part XIX Hazard Prevention Program 19.1